

FIELD MANUAL // v1.0 // 310+ RESOURCES

CALM AY PRESENTS

All-in-One Offensive Security Path

A curated, free-and-legal field manual: every platform, lab, cert, tool, repo, blog, channel and book — each with a link, what it is, and what it's best for. Built to take you from zero to dangerous.

```
techsis@offsec: ~/journey

$ whoami
  → future top-tier offensive security engineer
$ cat mission.txt
  "learn everything. pay for nothing but the exam."
$ ./start --path all-in-one # let's go
```

Curated for Esabu "Tech Sis" Blessing

Free & legal resources only · no leaked material

► 22 sections
recon → root → report

How to use this manual

This is a map, not a novel. You will not read it cover to cover. You will jump to a lane, grab the free lab and the free write-up, and go build the skill with your hands.

The core truth: almost everything taught in expensive courses is re-taught, free, by the community. Official cert links are included so you know the *target* — but the knowledge to get there is all in the free labs, repos, blogs and videos listed here.

01 Pick a lane, not everything

Foundations to Web to AD is the highest-ROI beginner path. Do not try to read this front to back.

02 Free-first

Every entry here has a free, legal way in. Official cert pages are listed so you know the target you are working toward, not because you must pay now.

03 Learn by doing

For every concept, find the matching free lab (HTB/THM/PortSwigger) and the matching write-up (lppSec/0xdf). Watch, then do it yourself with no hints.

04 Take notes obsessively

Keep your own methodology repo (Obsidian/Notion/GitHub). The notes you write beat any course PDF.

05 Certs = exam + paper, not secret knowledge

The knowledge in paid courses is re-taught free by the community. Pay for a cert when you need the credential, not to learn.

RULES OF ENGAGEMENT

Everything here is a legitimate, publicly-available learning resource — official sites, open-source repos, free labs, and creators' own published work. Only ever test systems you own or are explicitly authorized to test. Use isolated VMs for malware. Unauthorized access is a crime; the goal here is to become the professional who gets *hired* to break in.

Table of Contents

22 sections · 310+ hand-picked resources.

01 Training Companies & Academies

02 Certifications

03 Lab Platforms

04 Active Directory Labs

05 Web Security Labs

06 API Security

07 Mobile Security

08 Binary Exploitation (Pwn)

09 Reverse Engineering

10 Malware Analysis

11 Cloud Offensive Security

12 Wireless Security

13 IoT Security

14 Vulnerable Machines

15 Bug Bounty Platforms

16 Essential GitHub Repositories

17 Command & Control (C2) Frameworks

18 Research Blogs

19 Conferences (Free Talks on YouTube)

20 YouTube Channels

21 Books

22 The Tool Arsenal

23 Career & Learning Roadmap

01 // Training Companies & Academies

Where structured learning lives. Official pages for reference; the free tiers and community mirrors are where you actually start.

► Tier 1

OffSec

Knowing the industry-standard pentest path

The OSCP people. Courses are paid, but their free 'Proving Grounds Play' and the legendary PWK syllabus are public references.

offsec.com >

Hack The Box

Practical, gamified skill-building

Huge hands-on platform. Retired machines + free tier + HTB Academy 'Tier 0' modules are free.

hackthebox.com >

PortSwigger Web Security Academy

Web app security, start here for web

100% FREE. The best web-app hacking course on the internet, from the makers of Burp Suite.

portswigger.net >

SANS

Reference posters + cheat sheets

Elite (expensive) training. Their free 'Cyber Aces', posters, and cheat sheets are excellent references.

sans.org >

INE Security

Beginner-friendly structured pentest

Home of the eJPT. Frequently offers free eJPT-track content and a free starter pass.

ine.com >

TCM Security

Cheap + practical beginner path

Affordable, practical. Heath Adams' 'Practical Ethical Hacking' has huge free YouTube coverage.

academy.tcm-sec.com >

Zero-Point Security

Red team / Cobalt Strike tradecraft

Home of CRT0 (Red Team Ops). Note: new domain as of 2026. RastaMouse's free blog covers much of the tradecraft.

zeropointsecurity.co.uk >

Altered Security

Active Directory attack certs

Home of CRTP/CRTE (Active Directory red teaming). Nikhil Mittal's free 'Pentester Academy' talks cover the concepts.

alteredsecurity.com >

CyberWarFare Labs

Budget red-team certs

Cheap red-team certs (CRTA etc.). Frequent flash sales; free community labs and webinars.

cyberwarfare.live >

PentesterLab

Web + secure code review drills

Web + code review exercises. A rotating set of free exercises ('Essential' badge) is available.

pentesterlab.com >

► Tier 2

eLearnSecurity (now INE)

Recognized practical certs

Legacy brand folded into INE. Certs (eCPPT, eWPTX) still issued via INE.

ine.com >

EC-Council

HR-recognized baseline cert

Home of CEH. Widely known, HR-friendly. Concepts are heavily covered free elsewhere.

eccouncil.org >

Practical DevSecOps

DevSecOps specialization

DevSecOps + cloud-native security certs. Blog has free threat-modeling content.

practical-devsecops.com >

Antisyphon Training

Free/cheap high-quality courses

Pay-what-you-can model. Many genuinely FREE courses + Black Hills' free content firehose.

antisyphontraining.com >

AttackIQ Academy

MITRE ATT&CK + emulation, free

FREE courses on adversary emulation and MITRE ATT&CK.

academy.attackiq.com >

Immersive Labs

Enterprise cyber drills

Enterprise gamified labs. Occasionally free community challenges.

immersivelabs.com >

Security Blue Team

Defensive counterpart skills

Blue-side, but has a well-known FREE 'Intro to' course and BTL1 track.

securityblue.team >

Cybrary

Free intro breadth

Large free tier of intro cybersecurity courses.

cybrary.it >

OpenSecurityTraining2

Free deep low-level / RE courses

100% FREE, deep, university-grade courses on RE, exploitation, x86/x64, TrustZone.

ost2.fyi >

RangeForce

Defensive hands-on

Hands-on blue/purple modules. Community edition available.

rangeforce.com >

CyberDefenders

Understanding detection, free

FREE blue-team CTFs and DFIR labs (attacker's view for you).

cyberdefenders.org >

Pwned Labs

Free cloud pentesting labs

FREE cloud (AWS/Azure/GCP) attack labs. Excellent and current.

pwnedlabs.io >

Blue Team Labs Online

Seeing the defender's side

Free + paid defensive investigations.

blueteamlabs.online >

Red Team Village

Free red team community + talks

DEF CON village. Free talks, workshops, CTFs on YouTube.

redteamvillage.io >

02 //

Certifications

Official pages so you know the target. Remember: you pay for the exam and the paper. The knowledge is free below in every other section.

► OffSec

OSCP (PEN-200)

The pentest career door-opener

The industry-standard practical pentest cert. Free prep: TJnull's OSCP-like HTB/PG list + IppSec videos.

offsec.com >

OSEP (PEN-300)

Evasion / adversary simulation

Advanced evasion + AD. Free prep: 'Sektor7' free samples, Altered Security talks.

offsec.com >

OSWE (WEB-300)

White-box web / code review

White-box web exploitation. Free prep: PentesterLab + PortSwigger + source-code review practice.

offsec.com >**OSD (EXP-301)**

Windows exploit development

Windows exploit dev. Free prep: ROP Emporium, pwn.college, OpenSecurityTraining2.

offsec.com >**OSEE (EXP-401)**

Elite exploitation specialists

The hardest OffSec cert; advanced exploitation. Rare and elite.

offsec.com >**OSWP (PEN-210)**

Wireless pentesting

Wireless attacks. Free prep: WiFiChallenge Lab + Aircrack-ng docs.

offsec.com >**OSMR /
OSCC /
PEN-100 /
OSD**

Browsing the full OffSec ladder

Full OffSec catalog (macOS RE, cloud, foundations). Check the catalog for the current lineup.

offsec.com >**► Hack The Box****CPTS**

Modern OSCP alternative

Rapidly-respected practical pentest cert. Deep, modern, and cheaper than OSCP. Prep IS the Academy path.

academy.hackthebox.com >**CBBH**

Bug bounty methodology cert

Bug bounty focused. Strong web methodology.

academy.hackthebox.com >**CWEE**

Expert web exploitation

Advanced web exploitation.

academy.hackthebox.com >**CDSA**

Understanding detection

Blue-team analyst (know your enemy = the defender).

academy.hackthebox.com >**► TCM Security****PNPT**

Realistic, affordable pentest cert

5-day real-world pentest + AD, ending in a live report debrief. Beloved, affordable. Free prep: TCM's PEH on YouTube.

certifications.tcm-sec.com >**PJPT**

First hands-on AD cert

Junior AD-focused practical cert. Great confidence builder.

certifications.tcm-sec.com >**PEH (course)**

Free-heavy beginner course

The course, not a cert. Huge chunks are free on the TCM YouTube channel.

academy.tcm-sec.com >**► INE / eLearnSecurity****eJPT**

Absolute beginner practical cert

The best true-beginner practical cert. Cheap, entry-level. Free prep: INE's free eJPT track.

security.ine.com >**eCPPT**

Intermediate pentest cert

Mid-level practical pentester.

security.ine.com >

eWPT / eWPTX

Web-focused certs

Web app pentest (X = expert).

security.ine.com >

eCPTX

Advanced pentest cert

Expert-level pentest / red team.

security.ine.com >

► Zero-Point Security

CRTO

Red team operations cert

The go-to modern red-team cert (Cobalt Strike, AD).
Free prep: RastaMouse blog + GOAD lab.

zeropointsecurity.co.uk >

CRTL (RTO II)

Advanced red-team infra/lead

Red Team Lead, the advanced follow-on.

zeropointsecurity.co.uk >

CRTM

Offensive tooling development

Malware/offensive-dev focused track.

zeropointsecurity.co.uk >

► Altered Security

C RTP

Active Directory attacks, entry

The classic entry AD red-team cert. Free prep: GOAD lab + HackTricks AD + BloodHound docs.

alteredsecurity.com >

CRTE

Advanced AD / multi-forest

Multi-forest AD, the follow-on to C RTP.

alteredsecurity.com >

CARTP

Azure/Entra attacks

Azure AD (Entra) red teaming.

alteredsecurity.com >

CARTE / CTRL

Cloud red-team expert

Expert Azure + red-team lead tracks.

alteredsecurity.com >

► SANS (GIAC)

GPEN

Enterprise-recognized pentest

GIAC pentester. Expensive but HR-heavy.

giac.org >

GWAPT

Enterprise web pentest

GIAC web app pentester.

giac.org >

GXPEN

Advanced exploitation (GIAC)

Advanced pentest + exploit dev.

giac.org >

GREM

Malware reverse engineering

Malware RE. Free prep: OpenSecurityTraining2, MalwareUnicorn.

giac.org >

GMOB

Mobile pentest (GIAC)

Mobile security.

giac.org >

► EC-Council

CEH

Passing HR filters

The famous HR-checkbox cert. Broad but shallow; knowledge is all free elsewhere.

eccouncil.org >

CPENT / LPT

EC-Council practical track

EC-Council's hands-on pentest ladder.

eccouncil.org >

03 // Lab Platforms

Where you actually practice. Almost all have strong free tiers. This is where 80% of your hours should go.

Hack The Box (main)

Core practical practice

Active + retired machines. Retired boxes are free with a walkthrough by IppSec each.

hackthebox.com >

HTB Academy

Structured guided learning

Guided modules with theory + labs. 'Tier 0' modules are free.

academy.hackthebox.com >

HTB Pro Labs

Simulated full networks

Large multi-machine enterprise simulations (Dante, Zephyr, RastaLabs).

hackthebox.com >

HTB Sherlocks

Learning what defenders see

DFIR/blue investigations, some free.

hackthebox.com >

HTB Fortress / Battlegrounds

Competitive / themed challenges

Vendor-sponsored challenge sets + live PvP hacking.

hackthebox.com >

TryHackMe

Absolute beginners, start here

The most beginner-friendly platform. Huge free room library + guided learning paths.

tryhackme.com >

PortSwigger Academy

Web app hacking, free + best

FREE. The definitive web hacking labs.

portswigger.net >

PentesterLab

Web + secure code drills

Web + code review. Rotating free exercises.

pentesterlab.com >

Proving Grounds (OffSec)

OSCP-style box practice

PG Play is FREE (community VulnHub boxes); Practice is paid, OSCP-like.

offsec.com >

VulnHub

Free offline boxes

FREE downloadable vulnerable VMs to run offline. Classic OSCP prep.

vulnhub.com >

Root-Me

Broad free challenge bank

FREE, 400+ challenges across every category.

root-me.org >

OverTheWire

Linux + fundamentals, free

FREE wargames. 'Bandit' is the classic Linux/SSH starting point.

overthewire.org >

CyberDefenders

Detection perspective, free

FREE blue-team + DFIR labs.

cyberdefenders.org >

RangeForce

Blue-team hands-on

Hands-on defensive modules.

rangeforce.com >

MetaCTF

CTF practice

CTF platform + regular free competitions.

metactf.com >

PicoCTF

Beginner CTF, free

FREE, beginner CTF by CMU. Best on-ramp to CTFs.

picoctf.org >

Hacker101 CTF

Bug bounty on-ramp, free

FREE, by HackerOne. Earn private bug-bounty invites.

ctf.hacker101.com >

Google CTF

High-quality CTF challenges

FREE annual CTF; past challenges public on GitHub.

capturetheflag.withgoogle.com >

pwn.college

Binary exploitation, free + deep

FREE, from ASU. University-grade binary exploitation + more.

pwn.college >

pwnable.kr / pwnable.tw

Pwn practice, free

FREE binary exploitation wargames.

pwnable.kr >

Ringzer0 / SmashTheStack

Varied free challenges

FREE long-running challenge sites across many topics.

ringzer0ctf.com >

EchoCTF / ParrotCTF

Red-team style CTF

FREE red-team-flavored CTF environments.

echoctf.red >

04 // Active Directory Labs

AD is where most real-world compromise happens. These free labs teach the entire attack surface.

GOAD (Game of Active Directory)

Full AD attack practice, free

FREE, the gold-standard vulnerable AD lab. Multi-domain, deploy locally. THE way to practice AD attacks for free.

github.com >

DetectionLab

Attack + detection together

FREE AD lab wired with logging/EDR so you see attacks AND detections.

github.com >

PurpleSharp

Adversary simulation in AD

FREE tool to simulate AD attacks for detection testing.

github.com >

BloodHound (+ docs)

Understanding AD attack paths

FREE. The AD attack-path mapping tool; docs double as an AD attack course.

bloodhound.specterops.io >

HTB Pro Labs (AD)

Enterprise AD simulation

RastaLabs/Offshore/Cybernetics = enterprise AD simulations (paid).

hackthebox.com >

Altered Security / Zero-Point / CWL AD labs

Cert-aligned AD labs

The paid cert labs (CRTP/CRTE/CRTO). GOAD covers most concepts free first.

alteredsecurity.com >

05 // Web Security Labs

Deliberately vulnerable web apps. Free and endless. Pair with the PortSwigger Academy.

PortSwigger Web Security Academy

Web hacking, the #1 free resource

FREE. The single best resource for web hacking, period. Labs + theory for every vuln class.

portswigger.net >

OWASP Juice Shop

Modern web app practice, free

FREE modern vulnerable app (Node/Angular). Gamified with a scoreboard.

owasp.org >

DVWA

Beginner web vulns, free

FREE classic PHP vulnerable app with adjustable difficulty.

[github.com](#) >

bWAPP

Broad web vuln coverage

FREE app with 100+ vulnerabilities.

[itsecgames.com](#) >

Mutillidae II

OWASP Top 10 practice

FREE OWASP-mapped vulnerable app with hints.

[github.com](#) >

XVWA

Extra web vuln drills

FREE 'Xtreme' vulnerable web app.

[github.com](#) >

WebGoat / WebWolf

Guided web lessons, free

FREE OWASP teaching app (Java) with lessons.

[owasp.org](#) >

Google Gruyere

Guided web intro, free

FREE Google codelab-style vulnerable app.

[google-gruyere.appspot.com](#) >

NodeGoat / RailsGoat

Framework-specific vulns

FREE OWASP apps for Node.js / Rails specifics.

[github.com](#) >

Damn Vulnerable GraphQL App

GraphQL security, free

FREE, for learning GraphQL attacks.

[github.com](#) >

Damn Vulnerable REST API

REST API security

FREE, for REST API attack practice.

[github.com](#) >

OWASP SKF

Secure coding + labs

FREE secure-knowledge framework + labs.

[owasp.org](#) >

06 // API Security

APIs are the modern attack surface. All of these are free.

APISec University

API hacking course, free

FREE, full courses on API pentesting by Corey Ball.

[university.apisec.ai](#) >

crAPI

Hands-on API vulns, free

FREE OWASP 'completely ridiculous API' vulnerable lab.

[github.com](#) >

vAPI

OWASP API Top 10 practice

FREE vulnerable API mapped to OWASP API Top 10.

[github.com](#) >

DVGA (Damn Vulnerable GraphQL App)

GraphQL attacks

FREE GraphQL-specific target.

[github.com](#) >

PortSwigger API Labs

API testing, free + guided

FREE API testing labs within the Academy.

[portswigger.net](#) >

OWASP API Security Top 10

API risk reference

FREE the canonical reference for API risks.

[owasp.org](#) >

07 // Mobile Security

Android and iOS. The OWASP MASTG is your free bible here.

► Foundation

OWASP MASTG / MASVS

Mobile pentest bible, free

FREE. The Mobile App Security Testing Guide + standard. The definitive free mobile resource.

mas.owasp.org >

> Android

DIVA Android

Android vulns, free

FREE 'Damn Insecure and Vulnerable App' for Android.

github.com >

InsecureShop

Modern Android practice

FREE intentionally vulnerable Android app.

github.com >

DVAA

Android drills

FREE damn vulnerable Android application.

github.com >

Android Tamer

Android testing environment

FREE VM for Android security work.

androidtamer.com >

Frida Labs

Dynamic instrumentation, free

FREE labs to learn Frida instrumentation on Android.

github.com >

Oversecured Vulnerable App

Real-world Android vulns

FREE Android app packed with real vuln patterns.

github.com >

> iOS

iGoat

iOS lessons, free

FREE OWASP learning tool for iOS.

github.com >

DVIA-v2

iOS vulns, free

FREE Damn Vulnerable iOS App.

github.com >

Objection

Runtime mobile testing

FREE runtime mobile exploration (Frida-powered), iOS + Android.

github.com >

Needle (legacy)

iOS methodology reference

FREE (archived) iOS assessment framework, still educational.

github.com >

08 // Binary Exploitation (Pwn)

Memory corruption from zero. This is your hardware-adjacent lane, Esabu. All free.

pwn.college

Structured pwn learning, free

FREE, ASU university course. The best structured free path into pwn + RE.

pwn.college >

ROP Emporium

Learning ROP, free + focused

FREE, learn Return-Oriented Programming step by step across architectures.

ropemporium.com >

Exploit Education

Graduated exploit practice

FREE (Phoenix/Nebula/Protostar) graduated exploitation VMs.

exploit.education >

Microcorruption

Assembly + embedded, free

FREE embedded/assembly CTF in the browser. Addictive.

microcorruption.com >

pwnable.kr / pwnable.tw

Pwn challenges, free

FREE pwn wargames, easy to hard.

pwnable.kr >

HeapLAB (GEF/ how2heap)

Heap exploitation reference

FREE heap exploitation techniques catalog by Shellphish.

github.com >

CSAW / PicoCTF Pwn

Pwn CTF practice

FREE academic CTFs with strong pwn categories + public archives.

picoctf.org >

09 // Reverse Engineering

Understand binaries and malware internals. Ghidra makes the tooling free.

OpenSecurityTraining2

Deep free RE education

FREE deep RE + architecture courses (x86, ARM, exploitation).

ost2.fyi >

Malware Unicorn RE Workshops

Beginner RE, free + guided

FREE hands-on RE 101/102 workshops. Beginner favorite.

malwareunicorn.org >

Flare-On Challenges

RE skill-building, free

FREE annual Mandiant RE challenge; all past years public.

flare-on.com >

crackmes.one

RE practice bank, free

FREE huge archive of crackmes by difficulty.

crackmes.one >

Ghidra

Free industry-grade RE tool

FREE NSA-built decompiler/disassembler. Rivals IDA. Your default RE tool.

ghidra-sre.org >

REMnux

RE/malware toolkit distro

FREE Linux toolkit distro for RE + malware analysis.

remnux.org >

Malware Traffic Analysis

Network forensics practice

FREE real PCAP exercises with answers.

malware-traffic-analysis.net >

10 // Malware Analysis

Sandboxes and triage. Free tiers everywhere. (Handle real samples only in isolated VMs.)

ANY.RUN

Interactive triage, free tier

FREE interactive online sandbox (community tier). See malware behave live.

any.run >

Hybrid Analysis

Automated analysis, free

FREE automated malware sandbox + threat intel.

hybrid-analysis.com >

VirusTotal

Fast triage + intel, free

FREE multi-engine scan + rich intel (Behaviour tab is gold).

virustotal.com >

Joe Sandbox (Cloud Basic)

Deep sandbox, free tier

FREE community tier of a deep sandbox.

joesandbox.com >

CAPEv2

Self-hosted sandbox, free

FREE self-hosted sandbox specialized in config extraction.

github.com >

Cuckoo Sandbox

Open-source sandbox

FREE classic open-source automated analysis sandbox.

github.com >

PMA (Practical Malware Analysis labs)

Book-aligned malware labs

FREE lab binaries from the definitive book.

practicalmalwareanalysis.com >

11 // Cloud Offensive Security

AWS, Azure, GCP, containers. Pwned Labs + the *Goat family are the free core.

► Multi-cloud

Pwned Labs

Free cloud pentest labs, top pick

FREE, current, high-quality cloud attack labs across AWS/Azure/GCP.

pwnedlabs.io >

PortSwigger + HackTricks Cloud

Cloud attack reference, free

FREE HackTricks Cloud: attack methodology for every major provider.

cloud.hacktricks.wiki >

► AWS

flaws.cloud

AWS attacks intro, free

FREE guided AWS misconfiguration CTF. The classic starting point.

flaws.cloud >

flaws2.cloud

AWS attack + defense, free

FREE follow-up: attacker AND defender tracks.

flaws2.cloud >

CloudGoat

Scenario-based AWS, free

FREE 'vulnerable by design' AWS deployment tool by Rhino Security.

github.com >

► Azure

Altered Security Azure labs / HTB Academy

Azure cert labs

Paid cert labs; Pwned Labs covers Azure free first.

alteredsecurity.com >

MicroBurst / ROADtools

Azure tooling, free

FREE Azure attack tooling (NetSPI) + docs.

github.com >

► GCP

GCPGoat

GCP attacks, free

FREE vulnerable GCP deployment by INE.

[github.com](#) >

Containers / Kubernetes

Kubernetes Goat

K8s attacks, free + guided

FREE, the definitive vulnerable Kubernetes environment + guide.

[madhuakula.com](#) >

KubeGoat / kube-hunter

K8s recon, free

FREE K8s attack-surface hunter (Aqua).

[github.com](#) >

12 // Wireless Security

Wi-Fi attacks. The tooling is free; one great free lab exists.

WiFiChallenge Lab

Wireless practice, free + best

FREE (browser) full Wi-Fi pentesting lab. The best free way to practice wireless.

[wifichallengelab.com](#) >

Aircrack-ng

Wi-Fi cracking + docs, free

FREE the core Wi-Fi cracking suite; docs are a mini course.

[aircrack-ng.org](#) >

Bettercap

MITM + wireless recon

FREE Swiss-army knife for Wi-Fi/BLE/network MITM.

[bettercap.org](#) >

Kismet

Wireless recon/monitoring

FREE wireless detector/sniffer/IDS.

[kismetwireless.net](#) >

hcxumptool / hcxtools

PMKID / WPA capture, free

FREE modern WPA capture tooling (PMKID attacks).

[github.com](#) >

13 // IoT Security

Firmware and embedded. Overlaps your hardware interests directly.

IoTGoat

IoT firmware vulns, free

FREE OWASP deliberately-vulnerable IoT firmware.

[github.com](#) >

FirmAE

Emulating firmware, free

FREE firmware emulation framework (higher success rate).

[github.com](#) >

Firmadyne

Firmware emulation base

FREE the original firmware emulation system.

[github.com](#) >

CHIPSEC

Hardware/firmware auditing

FREE Intel platform/firmware security assessment framework.

[github.com](#) >

RouterSploit

Embedded exploitation, free

FREE exploitation framework for embedded devices.

[github.com](#) >

14 // Vulnerable Machines

Downloadable boxes to root offline. All free on VulnHub. Classic OSCP prep.

VulnHub (home of all below)

Free offline boxes, the hub

FREE. Download and root these VMs offline. Search any name below here.

[vulnhub.com](#) >

Kioptrix series

First boxes, free

FREE the classic beginner boot2root series.

[vulnhub.com](#) >

Metasploitable 2 & 3

Broad vuln practice, free

FREE intentionally vulnerable Linux/Windows targets by Rapid7.

[github.com](#) >

Mr-Robot / Raven / Stapler

Fun realistic boxes

FREE popular themed boot2root boxes.

[vulnhub.com](#) >

DC series / SickOS / NullByte

OSCP-style grind, free

FREE OSCP-like practice boxes.

[vulnhub.com](#) >

Brainpan

Intro to BOF, free

FREE the classic buffer-overflow box (OSCP legend).

[vulnhub.com](#) >

Symfonos / Sunset / VulnOS

Extra box practice

FREE more graded practice series.

[vulnhub.com](#) >

HackSys Extreme Vulnerable Driver

Kernel exploitation, free

FREE Windows kernel exploitation practice driver.

[github.com](#) >

15 // Bug Bounty Platforms

Where you get paid (and build reputation). Signing up is free everywhere.

HackerOne

Biggest program pool

The largest platform. Pair with the FREE Hacker101 CTF to earn private invites.

[hackerone.com](#) >

Bugcrowd

Programs + free training

Major platform + FREE 'Bugcrowd University' training.

[bugcrowd.com](#) >

Intigriti

Beginner-friendly bounties

Europe-strong platform, beginner-friendly, active community + free content.

[intigriti.com](#) >

YesWeHack + Dojo

Programs + free Dojo

Global platform. FREE 'Dojo' training arena.

[yeswehack.com](#) >

Synack Red Team

Elite vetted researchers

Vetted, invite-only, higher pay.

synack.com >

Open Bug Bounty

Coordinated disclosure, free

FREE coordinated-disclosure platform for web vulns.

openbugbounty.org >

Federacy

Smaller/startup scope

Bug bounty for startups.

federacy.com >

16 // Essential GitHub Repositories

The free backbone of day-to-day hacking. Star these. HackTricks alone replaces several paid courses.

HackTricks

The #1 free reference, all topics

FREE. The single most useful pentest wiki alive. Methodology for everything. Bookmark forever.

github.com >

PayloadsAllTheThings

Payloads/bypasses, essential

FREE payloads + bypasses for every vuln class. Your cheat-sheet vault.

github.com >

SecLists

Wordlists, essential

FREE the collection of wordlists (users, passwords, fuzzing, discovery).

github.com >

PEASS-ng (LinPEAS/WinPEAS)

Privesc enumeration, essential

FREE the go-to privilege-escalation enumeration scripts.

github.com >

GTFOBins

Linux privesc lookups

FREE Unix binaries abused to bypass restrictions / escalate.

gtfobins.github.io >

LOLBAS

Windows living-off-the-land

FREE the Windows equivalent of GTFOBins.

lolbas-project.github.io >

LOLDrivers

BYOVD reference

FREE living-off-the-land vulnerable drivers list.

loldrivers.io >

Priv2Admin

Windows privilege abuse

FREE Windows privilege-abuse escalation paths.

github.com >

HackTricks Cloud

Cloud methodology, free

FREE cloud + CI/CD attack methodology.

cloud.hacktricks.wiki >

linWinPwn / lse

AD automation, free

FREE automated AD enumeration/attack script.

github.com >

Atomic Red Team

ATT&CK technique practice

FREE small tests mapped to MITRE ATT&CK. Learn techniques atomically.

github.com >

PowerSploit / PowerView

AD PowerShell classics

FREE (archived) foundational AD/PowerShell offensive toolkit.

github.com >

16b // Command & Control (C2) Frameworks

Open-source C2s. Free to study and lab with. The core of red-team operations.

Sliver

Modern free C2, top pick

FREE, the modern open-source C2 by Bishop Fox. The current community favorite.

[github.com](#) >

Havoc

Modern free C2

FREE modern, sleek open-source C2.

[github.com](#) >

Mythic

Extensible C2 platform

FREE extensible, multi-agent C2 framework.

[github.com](#) >

Empire / Starkiller

Classic PowerShell C2

FREE PowerShell/Python post-exploitation C2.

[github.com](#) >

Covenant

.NET-based C2

FREE .NET C2 framework.

[github.com](#) >

PoshC2 / Merlin

Alternative free C2s

FREE additional open C2s (PowerShell / Go HTTP2).

[github.com](#) >

17 // Research Blogs

Where new techniques drop first. All free. Follow via RSS.

Google Project Zero

Cutting-edge vuln research

FREE elite 0-day research. Deep, humbling, essential reading.

[googleprojectzero.blogspot.com](#) >

PortSwigger Research

New web attack classes

FREE James Kettle & team invent whole new web attack classes here.

[portswigger.net](#) >

SpecterOps

Active Directory tradecraft

FREE the AD/BloodHound brain trust. Best AD tradecraft writing.

[posts.specterops.io](#) >

Trail of Bits

Systems + crypto research

FREE deep systems/crypto/blockchain security research.

[blog.trailofbits.com](#) >

Bishop Fox

Offensive tooling research

FREE offensive tooling + research (makers of Sliver).

[bishopfox.com](#) >

Synacktiv

Advanced exploitation

FREE elite exploitation write-ups (Pwn2Own winners).

[synacktiv.com](#) >

Black Hills Information Security

Practical pentest tips, free

FREE hugely practical, generous pentest content + webcasts.

[blackhillsinfosec.com](#) >

NCC Group Research

Broad deep research

FREE broad, high-quality security research.

[research.nccgroup.com](#) >

Oxdf (write-ups)

HTB write-ups, free + best

FREE the best HTB machine write-ups on the internet. Learn methodology.

[0xdf.gitlab.io](#) >

Orange Tsai

Elite web chains

FREE legendary web/SSRF/logic-chain research.

[blog.orange.tw](#) >

Securelist / Talos / Rapid7

Threat intel + advisories

FREE threat-intel + vuln research from major vendors.

securelist.com >

18 // Conferences (Free Talks on YouTube)

You do not need a ticket. Every talk below is free on YouTube. Gold mine.

DEF CON

Huge free talk archive

FREE thousands of talks on YouTube. The largest hacker con's full archive.

youtube.com >

Black Hat

Research briefings, free

FREE professional, research-heavy briefings on YouTube.

youtube.com >

BSides (various)

Community talks, free

FREE community cons worldwide, all recorded. Beginner-friendly.

youtube.com >

OffensiveCon

Exploitation research, free

FREE elite exploitation/vuln-research conference.

youtube.com >

Troopers

Deep technical, free

FREE deep technical talks (strong AD/network content).

youtube.com >

Nullcon / HITB / ShmooCon

More free talks

FREE more excellent recorded technical conferences.

youtube.com >

Red Team Village

Red-team workshops, free

FREE workshops + talks focused on offensive ops.

youtube.com >

19 // YouTube Channels

Free structured learning + walkthroughs. These are your daily teachers.

IppSec

HTB walkthroughs, essential

FREE, walks every retired HTB machine. THE channel for practical methodology.

youtube.com >

John Hammond

CTF + malware, beginner-friendly

FREE CTFs, malware analysis, and clear tutorials.

youtube.com >

LiveOverflow

Binary/pwn concepts, free

FREE deep binary exploitation + how-things-work explainers.

youtube.com >

The Cyber Mentor (Heath Adams)

Free full courses

FREE full ethical-hacking courses posted openly.

youtube.com >

Rana Khalil

Web methodology, free

FREE methodical PortSwigger Academy + web walkthroughs.

youtube.com >

NahamSec

Bug bounty, free

FREE bug bounty recon + real-world hunting.

youtube.com >

STÖK

Bounty mindset, free

FREE bug bounty mindset + hacker lifestyle/methodology.

youtube.com >

InsiderPhD

Beginner bug bounty, free

FREE the gentlest on-ramp to bug bounty + API hacking.

youtube.com >

PwnFunction

Visual web concepts, free

FREE beautifully animated web-vuln explainers.

youtube.com >

HackerSploit

Tooling tutorials, free

FREE broad pentest + Linux + tooling tutorials.

youtube.com >

NetworkChuck / David Bombal

Fundamentals + networking

FREE networking + Linux + cyber fundamentals.

youtube.com >

MalwareTech

Malware RE, free

FREE malware analysis + RE from the WannaCry-stopper.

youtube.com >

20 // Books

Foundational reading. Official/publisher links; several have free legal companion labs or drafts.

The Web Application Hacker's Handbook

Web hacking foundation

The web-hacking classic. Its free successor IS the PortSwigger Academy.

portswigger.net >

Real-World Bug Hunting

Bug bounty by example

Peter Yaworski's bounty book. His early drafts were community-shared; great real reports.

nostarch.com >

The Tangled Web

Browser security internals

Deep browser security model (Zalewski). Timeless.

nostarch.com >

Black Hat Python

Writing your own tools

Offensive tooling in Python. Code is free on the publisher's GitHub.

nostarch.com >

Linux Basics for Hackers

Linux for beginners

OccupyTheWeb's beginner Linux + hacking primer.

nostarch.com >

Practical Malware Analysis

Malware analysis foundation

The malware-analysis bible. Lab binaries are FREE online.

nostarch.com >

The Rootkit Arsenal

Rootkit internals

Deep Windows rootkit/anti-forensics internals.

google.com >

Red Team Development and Operations

Running red-team engagements

Companion site is FREE; the field guide to running red-team ops.

redteam.guide >

Metasploit: The Penetration Tester's Guide

Mastering Metasploit

The definitive Metasploit book (No Starch).

nostarch.com >

Practical Binary Analysis

Binary analysis foundation

Modern binary analysis + instrumentation.

nostarch.com >

Hacking: The Art of Exploitation

Exploitation from scratch

The classic that teaches exploitation from first principles. Includes a live CD.

nostarch.com >

21 // The Tool Arsenal

Every tool from your list, grouped by phase, each with its home + what it is best for. All free/open-source unless noted.

► Reconnaissance & Enumeration

Nmap

Port + service discovery

The port/host/service scanner. Learn it cold; NSE scripts extend it hugely.

nmap.org >

Masscan

Massive-scale scanning

Internet-scale asynchronous port scanner.

github.com >

RustScan

Fast port sweep

Ultra-fast port scanner that pipes into Nmap.

github.com >

Amass

Deep subdomain enum

OWASP subdomain enumeration + attack-surface mapping.

github.com >

Subfinder

Quick subdomain enum

Fast passive subdomain discovery (ProjectDiscovery).

github.com >

httpx

Probing live web hosts

Fast HTTP probing/fingerprinting toolkit.

github.com >

Katana

Crawling web apps

Next-gen crawler for endpoints/JS.

github.com >

naabu

Scriptable port scan

Fast port scanner (ProjectDiscovery).

github.com >

dnsx / shuffledns

DNS resolution/brute

Fast DNS toolkit + brute-forcing.

github.com >

gau / waybackurls

Historical URL discovery

Pull known URLs from Wayback/CommonCrawl.

github.com >

FFUF

Fuzzing everything web

Blazing-fast web fuzzer (dirs, params, vhosts).

github.com >

Gobuster

Content discovery

Directory/DNS/vhost brute-forcer.

github.com >

Feroxbuster

Recursive dir busting

Recursive content-discovery tool.

[github.com](#) >

dirsearch

Path discovery

Popular web path scanner.

[github.com](#) >

Arjun

Finding hidden params

HTTP parameter discovery.

[github.com](#) >

theHarvester

Passive OSINT recon

OSINT emails/subdomains/hosts gathering.

[github.com](#) >

SpiderFoot

Automated OSINT

Automated OSINT correlation engine.

[github.com](#) >

Maltego (CE)

OSINT link analysis

Link-analysis OSINT (free Community Edition).

[maltego.com](#) >

Recon-ng

Structured recon

Modular recon framework.

[github.com](#) >

Aquatone

Eyeballing attack surface

Visual recon (screenshots domains).

[github.com](#) >

► Web Application

Burp Suite (Community)

Web app testing, essential

The essential web proxy. Free Community edition + paid Pro. Non-negotiable tool.

[portswigger.net](#) >

OWASP ZAP

Free web proxy/scanner

FREE full-featured web proxy/scanner. The open Burp alternative.

[zapproxy.org](#) >

Caido

Lightweight modern proxy

Modern, fast web proxy (free tier). Rising Burp challenger.

[caido.io](#) >

SQLMap

SQL injection, essential

Automated SQL injection + DB takeover.

[sqlmap.org](#) >

XSSStrike

XSS hunting

Advanced XSS detection + fuzzing.

[github.com](#) >

Dalfox

Automated XSS

Fast parameter-based XSS scanner.

[github.com](#) >

Commix

Command injection

Automated command-injection exploitation.

[github.com](#) >

jwt_tool

Attacking JWTs

JWT testing/forging/attacks.

[github.com](#) >

GraphQLmap / Clairvoyance

GraphQL testing

GraphQL interaction + schema attacks.

[github.com](#) >

Kiterunner

API endpoint discovery

API content discovery (routes).

[github.com](#) >

► Active Directory

BloodHound / SharpHound

AD attack paths, essential

Maps AD attack paths visually. The single most important AD tool.

[github.com](#) >

Impacket

AD/Windows protocol attacks

The Python swiss-army suite for Windows protocols (secretsdump, psexec...).

[github.com](#) >

NetExec (nxc)

AD network operations

The maintained CrackMapExec successor. Network AD swiss-army knife.

[github.com](#) >

Certipy

ADCS attacks

AD Certificate Services (ADCS) enumeration + abuse.

[github.com](#) >

Rubeus

Kerberos attacks

Kerberos abuse toolkit (roasting, delegation, tickets).

[github.com](#) >

Mimikatz

Credential extraction

The legendary Windows credential-extraction tool.

[github.com](#) >

PowerView / SharpView

AD enumeration

AD enumeration from PowerShell.

[github.com](#) >

Seatbelt / SharpUp

Windows local recon

Host situational-awareness + privesc checks (GhostPack).

[github.com](#) >

Certify

ADCS (C#) attacks

C# ADCS abuse tool (pairs with Certipy).

[github.com](#) >

► Reverse Engineering & Debugging

Ghidra

Free RE, essential

FREE NSA decompiler. Your default RE platform.

[ghidra-sre.org](#) >

IDA Free

Industry-standard RE

Industry-standard disassembler (free tier; Pro is paid).

[hex-rays.com](#) >

Binary Ninja

Modern RE UX

Modern reversing platform (paid; free cloud demo).

[binary.ninja](#) >

Radare2 / Cutter

Free scriptable RE

FREE powerful RE framework (+ Cutter GUI).

[github.com](#) >

x64dbg

Windows debugging, free

FREE Windows user-mode debugger.

[x64dbg.com](#) >

dnSpy

.NET reversing, free

FREE .NET decompiler/debugger/editor.

[github.com](#) >

GDB + GEF/pwndbg

Linux exploit debugging

FREE enhanced GDB for exploitation.

[github.com](#) >

► Mobile

MobSF

Mobile analysis, essential

FREE automated static+dynamic mobile analysis. Start every mobile test here.

[github.com](#) >

Frida

Runtime hooking, essential

FREE dynamic instrumentation toolkit. Hook anything at runtime.

[frida.re](#) >

Objection

Runtime mobile testing

FREE Frida-powered runtime mobile exploration.

[github.com](#) >

JADX

Reading APKs, free

FREE Android APK to readable Java decompiler.

[github.com](#) >

apktool

APK repackaging

FREE decode/rebuild APK resources.

[github.com](#) >

Drozer

Android IPC attacks

FREE Android attack-surface assessment framework.

[github.com](#) >

► Malware Tooling

YARA

Malware detection rules

FREE pattern-matching for malware classification.

[github.com](#) >

capa

Triaging binaries

FREE identifies capabilities in executables.

[github.com](#) >

FLOSS

Deobfuscating strings

FREE extracts obfuscated strings from malware.

[github.com](#) >

Detect It Easy (DIE)

Packer detection

FREE packer/compiler identification.

[github.com](#) >

PE Studio / PE-bear

PE header analysis

FREE PE-file inspection.

[github.com](#) >

REMnux

Malware toolkit distro

FREE the malware-analysis toolkit distro.

[remnux.org](#) >

► Cloud & Container

Pacu

AWS attacks, essential

FREE the AWS exploitation framework ('Metasploit for AWS').

[github.com](#) >

ScoutSuite

Cloud misconfig audit

FREE multi-cloud security auditing.

[github.com](#) >

Prowler

Cloud posture assessment

FREE AWS/Azure/GCP security assessments + CIS.

[github.com](#) >

Trivy

Container scanning

FREE container/IaC/vuln scanner (Aqua).

[github.com](#) >

kube-hunter / kube-bench

K8s security checks

FREE Kubernetes hunting + CIS benchmarking.

[github.com](#) >

► Wireless

Aircrack-ng

Wi-Fi cracking

FREE the core Wi-Fi cracking suite.

aircrack-ng.org >

Bettercap

Wireless MITM

FREE network/wireless/BLE MITM framework.

bettercap.org >

Kismet

Wireless monitoring

FREE wireless sniffer/IDS.

kismetwireless.net >

hcxdumptool

WPA capture

FREE modern WPA/PMKID capture.

github.com >

► Password Attacks

Hashcat

GPU hash cracking, essential

FREE the world's fastest GPU password cracker.

hashcat.net >

John the Ripper

Versatile cracking

FREE versatile CPU-first cracker (great with 'jumbo').

openwall.com >

Hydra

Online brute-force

FREE fast network login brute-forcer.

github.com >

Medusa

Parallel brute-force

FREE parallel network brute-forcer.

github.com >

CeWL

Targeted wordlists

FREE spiders a site to build custom wordlists.

github.com >

► Red Team / C2 (see 16b)

Sliver / Havoc / Mythic

Free C2 operations

FREE modern open-source C2s. See section 16b for the full lineup.

github.com >

Empire / Covenant / PoshC2 / Merlin

Alternative C2s

FREE additional open C2 frameworks.

github.com >

Metasploit Framework

Exploitation framework, essential

FREE the foundational exploitation framework. Learn it early.

github.com >

22 // Career & Learning Roadmap

Turning all of the above into a job. Free roadmaps + how to build proof.

The Beginner Path (free)

Your first 6 months

Foundations (TryHackMe) to Web (PortSwigger) to Boxes (HTB retired + IppSec) to eJPT/CPTS. Do this in order.

tryhackme.com >

TJnull's OSCP Prep List

Structured OSCP prep, free

FREE the community list of OSCP-like HTB + PG boxes.

docs.google.com >

HackTricks (methodology)

Day-to-day methodology

FREE use as your always-open methodology reference during every lab.

book.hacktricks.wiki >

Build a notes/ methodology repo

Proof of skill + retention

FREE. Document every box + technique in Obsidian/ GitHub. This becomes your portfolio + brain.

obsidian.md >

Bug bounty as proof

Public track record

FREE public reports (Hacktivity) teach real bugs + build your reputation.

hackerone.com >

Open-source contributions

Credibility + networking

Contribute to the free tools above. Visible, credible, and teaches internals.

github.com >

Roles to target

Understanding the ladder

Penetration Tester to Red Teamer to Security Researcher / Exploit Dev. Map certs to each.

offsec.com >

Now go get **root**.

You have the entire map in your hands. The only thing between you and world-class is reps — boxes rooted, write-ups read, notes written, bugs reported. Pick section 01, open a lab, and start today. Then do it again tomorrow.

Foundations → Web → Active Directory → pick your specialty. One box a day. One technique at a time.

```
$ echo "built for Tech Sis – from recon to NVIDIA" ✓  
Calm Ay // All-in-One Offensive Security Path // stay curious, stay legal.
```